CIE STYLE

OPBP Physics Question Bank

CANDIDATE NAME

CLASS / CENTRE

CANDIDATE NUMBER

PHYSICS

OPBP / MOCK

Multiple Choice and Structured Questions

MOCK EXAMINATION

18 minutes

OPBP物理题库 ▲ Dynamics专项练习

INSTRUCTIONS

Attempt every question in the given order.

A calculator may be used.

Show formulas and calculations for numerical parts.

Write explanations and draw labelled diagrams in the spaces provided.

Use the parameters stated in each source question.

INFORMATION

6 source questions; total available marks: 12.

Marks for individual parts appear in square brackets.

Questions are sourced from OPBP. Mark allocation uses teacher rules.

This is a CIE-style mock paper.

Data and formulae

acceleration of free fall	g = 9.81 m/s^2
speed of light in vacuum	c = 3.00 x 10^8 m/s
elementary charge	e = 1.60 x 10^-19 C
electron mass	m_e = 9.11 x 10^-31 kg
proton mass	m_p = 1.67 x 10^-27 kg
gravitational constant	G = 6.67 x 10^-11 N m^2/kg^2
Coulomb constant	k_e = 8.99 x 10^9 N m^2/C^2
Planck constant	h = 6.63 x 10^-34 J s

$s = ut + (1/2)at^2$

$v^2 = u^2 + 2as$

$F = ma$; $p = mv$; impulse = change in momentum

$W = Fd\cos(\theta)$; $E_k = (1/2)mv^2$

$P = E/t$; $V = IR$; $Q = It$

$v = f\lambda$; $T = 1/f$

Values explicitly given in a source question take priority over this data sheet.

Multiple Choice

1 **P402 - Falling Mass on Rope** [Total: 1]

Part 1 [1]

C The change in momentum of a falling mass attached to a fixed rope is equal to:

A. $mg(t_2 - t_1)$ ✓

B. $mv^2/2$ ✗

C. integral of the net force over the stretching time of the rope

D. mass multiplied by the average velocity

E. $mg(L_2 - L_1)$ ✗

$mv = Fot$.



2 **P403-1 - Force Multipart True False Question** [Total: 1]

A Determine whether the following statements are true or false.

Part 1 [1]

An object undergoing uniform circular motion has a constant angular velocity.

A. True

B. False

3 **P478 - Position Vector and Momentum** [Total: 1]

Part 1 [1]

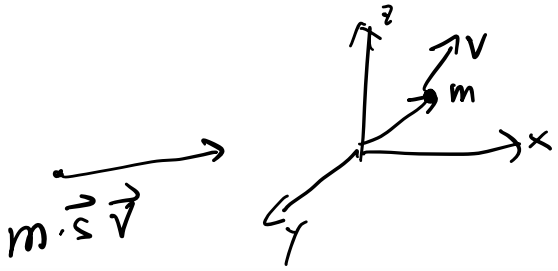
The cross product of the position vector with mass mutilpied with the velocity vector is the

A. Linear momentum

B. Angular momentum

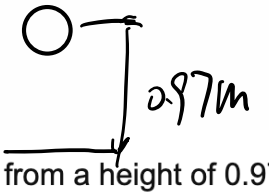
C. Kinetic Energy

D. Impulse



Structured Questions

4 **P087 - Bouncing Ball** [Total: 3]



A ball of mass 0.354 kg is dropped vertically from a height of 0.97 m and bounces back to the original height.

Part 1 [3]

What is the magnitude of the impulse that the floor exerts on the ball during the bounce?

Please enter in a numeric value in kg m/s.

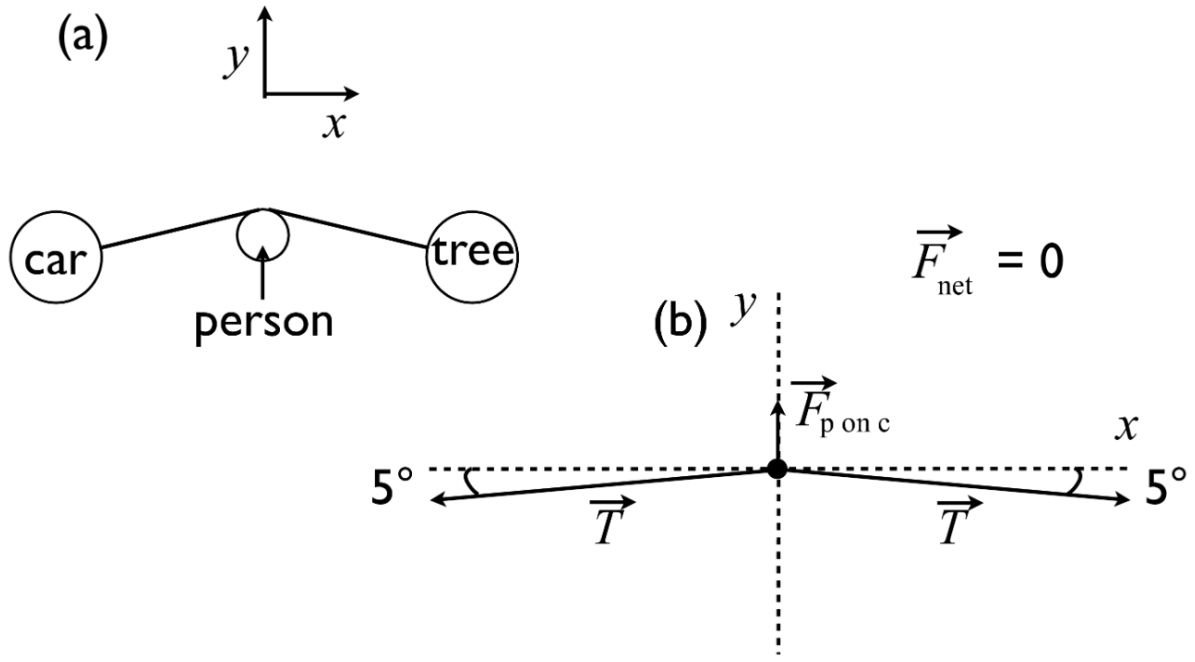
$v^2 = 2gh$ $v^2 = 2 \times 9.81 \times 0.97 = 4.36$ m/s

$I = \Delta p = mv$ $I = 0.354 \times 2 \times 4.36$

$= 3.09$ kg m/s

5 **P352 - Car in Ditch** [Total: 3]

One way to move a heavy object (like a car) is to attach a strong chain between it and a strong tree and push perpendicular to the chain (see Fig. (a)) to create a large tension. A free body diagram of a piece of the chain is shown in Fig. (b).



Part 1 [3]

If the person pushes on the chain with a force of 2000 N, the tension in the chain is:

Please enter in a numeric value in N.

$\vec{F}_{p\ on\ c} = 2 \cdot T \sin 5°$

$2000 = 2 \times T \cdot \sin 5°$

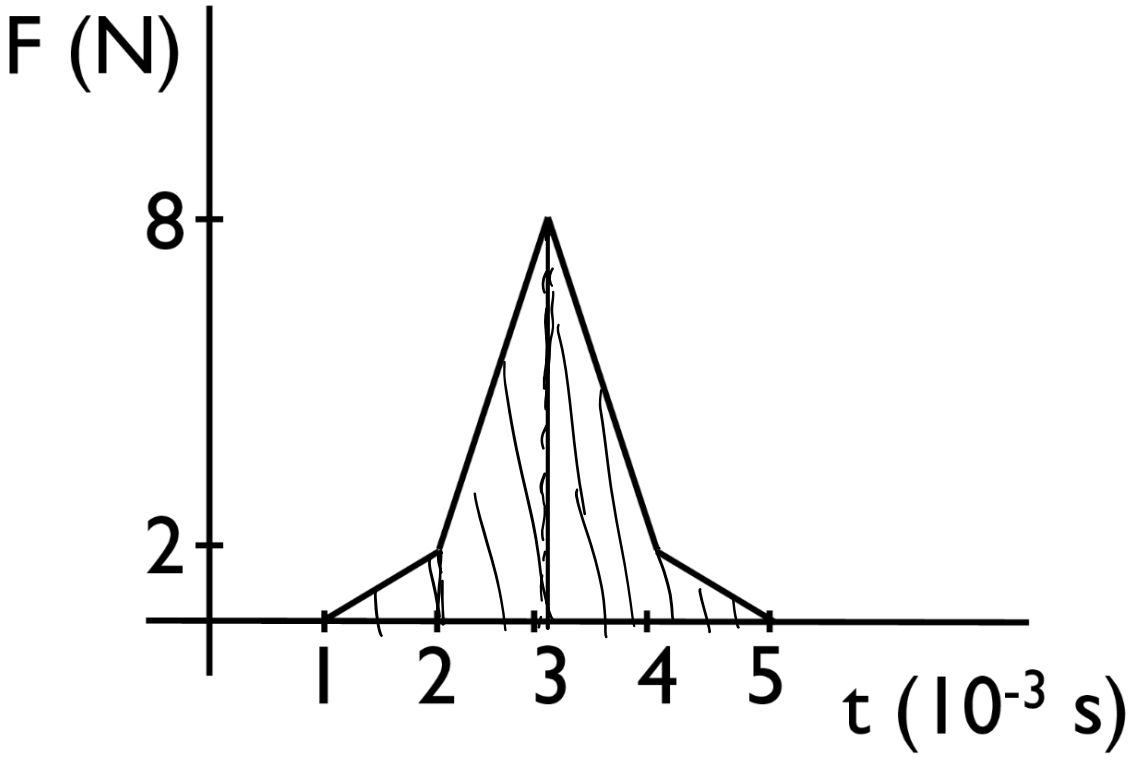
$T = 11474\ N$

6 **P456 - Force on a Mosquito** [Total: 3]

Part 1 [3]

A force versus time graph for the force on a mosquito is given in the figure below.

Calculate the change in momentum of the mosquito due to this impulsive force.



Please enter in a numeric value in $\frac{kgm}{s}$.

change in momentum is the shaded area

area $= [\frac{1}{2} \times 1 \times 2 + \frac{1}{2}(2+8) \times 1] \times 2 \times 10^{-3}$

$= 1.2 \times 10^{-2}$ kg m s^{-1}